



Nebraska Public Power District

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10 CFR 50.73

NLS2020066
December 30, 2020

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2020-004-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2020-004-00.

There are no regulatory commitments contained in this request.

Sincerely,

John Dent, Jr.
Vice President and
Chief Nuclear Officer

/jo

Attachment: Licensee Event Report 2020-004-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via IRIS entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

IEZ2
NRR



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R 3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/rs1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk for ora_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Cooper Nuclear Station	2. Docket Number 05000298	3. Page 1 OF 4
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4. Title Secondary Containment Differential Pressure Perturbation Could Have Prevented Fulfillment of a Safety Function
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
11	03	2020	2020	- 004 -	00	12	30	2020	Facility Name	Docket Number
										05000
										05000

9. Operating Mode	2	10. Power Level	3
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)				
10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(II)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(I)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ Other (Specify here, in Abstract, or in NRC 366A)				

12. Licensee Contact for this LER

Licensee Contact

Linda Dewhirst, Regulatory Affairs and Compliance Manager

Phone Number (Include Area Code)

(402) 825-5416

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
D	NG	CDMP		Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On November 3, 2020, Secondary Containment differential pressure (d/p) exceeded the Technical Specifications limit of -0.25 inches of water gauge (wg) causing Secondary Containment to be inoperable for 27 seconds.

Two causal factors were identified for this event. The first being that the manual pressure regulator, a non-essential component, drifted causing an un-demanded increase in Reactor Building HVAC supply flow. The second factor is a lack of procedural guidance to optimize the tuning parameters of the reactor building differential pressure controllers to maintain the required d/p during changing conditions.

The pressure regulator was replaced. Adoption of Technical Specification Task Force Traveler 551, Revise Secondary Containment Surveillance Requirements, will be pursued to eliminate the reporting requirements from exceeding -0.25 inches wg due to a failure of non-safety related equipment. Procedures will be updated to provide guidance to optimally tune the fans.

There was no impact on nuclear safety, plant reliability, radiological safety, or industrial safety.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Cooper Nuclear Station	05000-298	2020	- 004	- 00

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station was in Mode 2, Startup, at 3 percent power at the time of the event on November 3, 2020.

BACKGROUND

Secondary Containment serves as a barrier to confine and monitor potential releases during fuel handling operations and is a system that limits the release of radioactive materials to the environment and consists of four subsystems. One of the subsystems is the Reactor Building [EIS: NG], which encloses the Reactor Pressure Vessel [EIS: RPV] and Primary Containment [EIS: NH].

The secondary containment area (most of the Reactor Building) has supply and exhaust ventilating systems. The supply system furnishes filtered 100% outdoor air to all floors of the building through ductwork. The supply unit has an operating and standby (100% capacity) fan with vortex dampers [CDMP] to provide regulation of air capacity. The fans will deenergize in the event of loss of offsite power. The exhaust air is induced from the ventilated areas to a common plenum connected to the two exhaust fans, each of 100% capacity. The air is then exhausted to the atmosphere.

During normal plant operation, a minimum average negative pressure of 0.25 inches of water gauge (wg) is maintained by differential pressure controllers which receive signals proportional to the pressure difference between outside air and the secondary containment atmosphere, and control the position of the exhaust fan vortex dampers. The differential pressure is detected by four separate probes, with one on each side of the Reactor Building. Control action is initiated from the average value of the four sensor probes.

If a Loss of Coolant Accident should occur, all ventilation systems of the primary containment area and secondary containment area will be isolated automatically, and the Standby Gas Treatment system will be automatically initiated.

EVENT DESCRIPTION

On November 3, 2020, at 08:14, Reactor Building differential pressure (d/p) exceeded the Technical Specification limit of -0.25 inches of water gauge (wg) causing Secondary Containment to be inoperable for 27 seconds. Reactor Building HVAC Exhaust Fan 'B' (HV-FAN-EF-R-1B) was operating with Reactor Building Differential Pressure Controller 'B' (HV-DPIC-835B) in automatic at a differential pressure at approximately -0.34 inches wg. HV-DPIC-835B responded by raising output to restore d/p to -0.34 inches wg.

A review of operating logs identified that on October 12, 2020, HV-FAN-SF-R-1A-A was placed in service and remained in operation through-out the event. Flow is modulated through the fan by changing the throttled position of the inlet vortex dampers, HV-VD-VD1005A and HV-VD-VD1005A1. The position of these vortex dampers is controlled by a 3-18 psig manual pressure regulator (HV-FCV-1013). This regulator provides an output pneumatic air pressure signal to the vortex damper actuator control valve positioners. If more air flow is necessary, the output from the regulator is raised, increasing air flow through the in-service Reactor Building supply fan. Reactor building differential pressure will rise as a result of the higher influent flow. HV-DPIC-

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835A/B will sense the rising pressure and raise the Reactor Building exhaust air flow to restore the desired building differential pressure.

When the air flow rose, HV-DPIC-835B did respond to the changing flow, however the controller lagged in response, and the reactor building differential pressure exceeded -0.25 inches wg prior to increasing exhaust flow.

The manual pressure regulator is a non-essential component associated with the non-safety related reactor building HVAC system. The industry has recognized a loss of these non-safety related components does not affect the ability of secondary containment to perform its required safety function. Technical Specification Task Force Traveler (TSTF) 551, Revise Secondary Containment Surveillance Requirements, has been issued to recommend changes to Technical Specifications to eliminate the reporting requirements from exceeding -0.25 inches wg due to a failure of non-safety related equipment.

Although the differential pressure controller did respond, it was sluggish in its response. The tuning parameters are adjusted per procedure. A controller auto-tune program is run which cycles the reactor building differential pressure. The controller monitors system response, and tuning parameters are implemented. If the reactor building differential pressure is stable after these tuning parameters are used, no further adjustments are made. The procedure does not ensure the controller will be responsive enough to maintain reactor building d/p within limits during a transient. The guidance is subjective and does not challenge the controller to an upset condition to monitor response.

BASIS FOR REPORT

This event is reportable under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident. The secondary containment was operable at the time of discovery; therefore, it was not reportable under 10 CFR 50.72(b)(3)(v) as an 8 Hr ENS Notification.

An engineering review demonstrated that the perturbation on Secondary Containment pressure caused by the non-essential Reactor Building HVAC system did not impact the integrity of Secondary Containment and had no impact on the Standby Gas Treatment system. In the event of an accident, the Standby Gas Treatment system would have automatically started and would have been able to maintain vacuum on Secondary Containment as assumed in the Loss of Coolant Accident dose analysis.

Based on the engineering review, this event will not be counted as a Safety System Functional Failure for the Nuclear Regulatory Commission performance indicator since no loss of safety function occurred.

SAFETY SIGNIFICANCE

This condition was non-consequential. The pressure excursion in the reactor building resulted in an unexpected plant condition and met the entry condition for required action pursuant to Technical Specifications. This event resulted in minor excursion of Reactor Building d/p above the Technical Specification limit of -0.25 inches wg for 27 seconds until the building HVAC controls restored the required differential pressure. This event did not negate the ability of the Standby Gas Treatment System to maintain



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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		2020	- 004	- 00

Secondary Containment differential pressure, if required. There was not an impact on nuclear safety, plant reliability, radiological safety, or industrial safety.

CAUSE

Two causal factors were identified for this event. The first being that the manual pressure regulator, a non-essential component drifted causing an un-demanded increase in Reactor Building HVAC supply flow. The second factor is a lack of procedural guidance to optimize the tuning parameters of the reactor building differential pressure controllers to maintain the required d/p during changing conditions.

CORRECTIVE ACTIONS

The regulator (HV-FCV-1013) was replaced. Adoption of TSTF-551 will be pursued; additionally procedures will be updated to provide guidance to optimally tune the fans.

PREVIOUS EVENTS

On August 6, 2020, Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2020-002-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On May 1, 2020, Secondary Containment was breached due to both airlock doors in the Reactor Building being inadvertently opened simultaneously. This was reported under LER 2020-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On August 8, 2019, Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2019-002-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On January 6, 2014, Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2014-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On September 10, 2012, Secondary Containment was breached due to both airlock doors in the Reactor Building being inadvertently opened simultaneously. This was reported under LER 2012-003-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) - An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.